

CE_Player pipeline on video1 pictures - run report

Repo: jekeymer/terraquest-mint, cv/realrun. Frames: video1/pictures image0009 -> image0020 (12 frames, 3280x2464, Feb 20 2020). Delta t = 2 minutes per frame (Informe 1). Rendered by committed CI scripts, regenerates on every CI run.

```
pipeline_version = cv-pipeline v0.2 + ce_player_video1 v0.1 (CE_Player params verbatim)
score_formula = PENDING (this run validates tracked centers, not the u16 score)
delta_t = 2 minutes per frame (Informe 1 verbatim; notebook factor 100000/(2*scale*scale_percent), the 2 = time
step)
frame_order = sorted(glob('./pictures/*.jpg')) -> image0009..image0020, 12 frames
```

CE_Player video1 validation run

Run status

****Complete - real data, CI-computed, no fabricated values.**** Workflow
`realrun-video1` run 1 concluded ****success****; the runner executed end to end
and committed `cv/realrun/ce\_player\_video1\_results.json` and
`cv/realrun/ce\_player\_video1\_run\_log.txt` at commit `14eb48b`. Every number
below comes from that run's log.

Path of execution: my compute sandbox had no network (DNS failures), so the
run moved into GitHub Actions, which downloads the 12 frames + npz from
Suzana's shared Drive, runs both stages, and commits the results back to this
directory. CI-authored commits use GITHUB_TOKEN and cannot retrigger
workflows, so there is no commit loop.

Input verification evidence

- 12 frames, byte sizes 4,259,183 - 4,361,970 (contract 3.0-5.5 MB), all
decode 3280x2464, order image0009 -> image0020 verified against the
`sorted(glob)` convention.
- `data.npz`: 456 bytes, key `arr\_0`, shape (12, 2), float64, value range
[158.334274, 938.5].

Coordinate space

****40% resize (1312x986) - npz space, velocity_px_scale = 1.0.****
Evidence: TLD centers span x = [262.0, 938.5], y = [158.334, 887.0] - all
below the 1400/1000 bounds, matching her crop + resize-40% pipeline, so the
notebook factor applies directly to processing pixels.

Stage 1 - Lucas-Kanade optical flow

CE_Player params verbatim: `goodFeaturesToTrack(qualityLevel=0.01,
minDistance=4)` + 10px grid thinning; `calcOpticalFlowPyrLK(winSize=(15,15),
maxLevel=2, criteria=(EPS|COUNT,10,0.01))`; forward-backward rejection at
0.5 px.

Pair	Frames	Grid points	FB kept	Mean disp (px)
---:	---	---:	---:	---:
0	0009->0010	465	167	2.293409
1	0010->0011	650	163	0.871458
2	0011->0012	645	123	1.142603
3	0012->0013	705	160	1.518169
4	0013->0014	705	176	1.360365
5	0014->0015	870	256	0.880256
6	0015->0016	1029	358	1.418972
7	0016->0017	1055	374	2.154039
8	0017->0018	1205	444	2.111675
9	0018->0019	1587	341	4.765700
10	0019->0020	1546	592	1.627912

LK mean = 1.635 px, std(ddof=0) = 0.926 px over the 11 pairs. FB rejection
is heavy (123-592 of 465-1587 survivors) - see observations.

Stage 2 - frame-diff blob

absdiff -> GaussianBlur(5,5) -> Otsu -> open 3x3 -> largest-contour centroid,
assigned to frame i+1.

Frame	Center (px, 40% space)	Contour area (px^2)	Error vs TLD (px)
1	(720.888, 191.571)	7272.0	26.093
2	(742.252, 262.630)	7392.5	40.357
3	(769.509, 327.253)	7869.0	55.542

CE_Player diff view: white clusters, boxes, stats block

CE_Player output - video1 [its own DIFFBLOB view: absdiff -> blur 11 -> x3 -> thr 30 (ofApp.cpp on_Simpleblob)

white = thresholded pixel clusters | blue rect = [x1][y1][x2][y2] | [S] = area stats = blob | (red = follower trail space: 40% (1312x786), dt = 2 min
blob track v = 65.56 um/min vs Suzana TLD 74.1 +/- 12.2 (0.7 sigma) | mean err 127.5px | clusters below chfinder min area show white but carry no stats block



[x] 0
[y] 0
[x1] 0.0 [y1] 0.0
[S] 3196 [area] 0.04



[x] 2
[y] 564 [y1] 0
[x1] 0.0 [y1] 0.0
[S] 34294 [area] 0.19



[x] 3
[y] 560 [y1] 0
[x1] 0.0 [y1] 0.0
[S] 34324 [area] 0.79

1/12 image0009 (base frame - diff starts on the next frame)



[x] 3
[y] 575 [y1] 114
[x1] 0.0 [y1] 0.0
[S] 35226 [area] 0.36

2/12 image0010.jpg <- image0009.jpg blob=miniArea 1



[x] 3
[y] 592 [y1] 171
[x1] 0.0 [y1] 0.0
[S] 36747 [area] 0.73

3/12 image0011.jpg <- image0010.jpg blob=miniArea 3



[x] 4
[y] 612 [y1] 214
[x1] 0.0 [y1] 0.0
[S] 38393 [area] 0.98

4/12 image0012.jpg <- image0011.jpg blob=miniArea 2



[x] 6
[y] 665 [y1] 344
[x1] 0.0 [y1] 0.0
[S] 38868 [area] 0.45

5/12 image0013.jpg <- image0012.jpg blob=miniArea 1



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

6/12 image0014.jpg <- image0013.jpg blob=miniArea 1



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

7/12 image0015.jpg <- image0014.jpg blob=miniArea 2



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

8/12 image0016.jpg <- image0015.jpg blob=miniArea 3



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

9/12 image0017.jpg <- image0016.jpg blob=miniArea 2



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

10/12 image0018.jpg <- image0017.jpg blob=miniArea 3



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

11/12 image0019.jpg <- image0018.jpg blob=miniArea 6



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

12/12 image0020.jpg <- image0019.jpg blob=miniArea 4



[x] 6
[y] 686 [y1] 327
[x1] 0.0 [y1] 0.0
[S] 38870 [area] 0.61

Track comparison: CE_Player vs TLD ground truth

CE_Player output - video1 | 12 frames Feb 20 2020 | dt = 2 min | space 40% (1312x986)
red = our frame-diff blob track green = Suzana TLD (data.npz) thumbs 0.196x
v = 65.56 um/min vs TLD 74.1 +/- 12.2 (0.7 sigma) | mean err 127.5px | max 778px (last frame)

